1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
d5_96305_sanchit> delimiter $$
d5_96305_sanchit> create procedure sp_rectangle(in p_l int, in p_b int, out p_perimeter int , out p_area int)
    -> begin
    -> set p_perimeter = 2*(p_l + p_b);
    -> set p_area = p_l * p_b;
    -> select p_l as length, p_b as breadth, p_perimeter as perimeter, p_area as area;
    -> end;$$
Query OK, 0 rows affected (0.348 sec)

d5_96305_sanchit> call sp_rectangle(5,10,@a,@b);$$
+--------+---------+-----------+------+
| length | breadth | perimeter | area |
+--------+---------+-----------+------+
|      5 |      10 |        30 |   50 |
+--------+---------+-----------+------+
1 row in set (0.011 sec)

Query OK, 0 rows affected (0.040 sec)

d5_96305_sanchit> |
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
d5_96305_sanchit> create procedure sp_power(p_num int)
    -> begin
    -> declare v_num int;
    -> declare v_square int;
    -> declare v_cube int;
    -> set v_num = p_num;
    -> set v_square = p_num * p_num;
    -> set v_cube = power(p_num, 3);
    -> insert into test values(v_num,v_square,v_cube);
    -> end;$$
Query OK, 0 rows affected (0.229 sec)

d5_96305_sanchit> call sp_power(5);
    -> $$
Query OK, 1 row affected (0.207 sec)

d5_96305_sanchit> select * from test;$$
+-------+---------+-------+
| numbr | squares | cubes |
+-------+---------+-------+
|     5 |      25 |   125 |
+-------+---------+-------+
1 row in set (0.006 sec)

d5_96305_sanchit> delimiter ;
d5_96305_sanchit> |
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
d5_96305_sanchit> delimiter $$
d5_96305_sanchit> create procedure sp_temp_conv(in p_c int, in p_f int, out p_fahreniet double, out p_celcius double)
    -> begin
    -> set p_fahreniet = (9/5) * p_c + 32;
    -> set p_celcius = (p_f - 32) * (5/9);
    -> end;
    -> $$
Query OK, 0 rows affected (0.261 sec)

d5_96305_sanchit> call sp_temp_conv(99,380, @a,@b);
    -> $$
Query OK, 0 rows affected (0.159 sec)

d5_96305_sanchit> select @a as temp_in_f, @b as temp_in_c; $$
+-----------+-----------+
| temp_in_f | temp_in_c |
+-----------+-----------+
|     210.2 |  193.3488 |
+-----------+-----------+
1 row in set (0.057 sec)

d5_96305_sanchit> |
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d5_96305_sanchit> create procedure metric(p_inches int)
    ->     begin
    ->     declare v_yard int;
    ->     declare v_foot int;
    ->     declare v_inch int;
    ->     set v_yard = p_inches div 36;
    ->     set v_foot = (p_inches mod 36) div 12;
    ->     set v_inch = (p_inches mod 36) mod 12;
    ->     select p_inches, concat(v_yard," ", "yards ",v_foot," foots ",v_inch," inches");
    ->     end;
    ->     $$
Query OK, 0 rows affected (0.264 sec)

d5_96305_sanchit> call metric(124);
    -> $$
+----------+-------------------------------------------------------------------+
| p_inches | concat(v_yard," ", "yards ",v_foot," foots ",v_inch," inches") |
+----------+-------------------------------------------------------------------+
|      124 | 3 yards 1 foots 4 inches                                          |
+----------+-------------------------------------------------------------------+
1 row in set (0.020 sec)

Query OK, 0 rows affected (0.066 sec)

d5_96305_sanchit> |
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
d5_96305_sanchit> create procedure greater_than_hundred(num1 int, num2 int)
    ->     begin
    ->     declare v_product int;
    ->     set v_product = num1 * num2;
    ->     if v_product > 100 then select ' product is greater than 100';
    ->     else select 'product is less than 100';
    ->     end if;
    ->     end;$$
Query OK, 0 rows affected (0.410 sec)

d5_96305_sanchit> call greater_than_hundred(20,6);$$
+------------------------------+
| product is greater than 100  |
+------------------------------+
|  product is greater than 100 |
+------------------------------+
1 row in set (0.013 sec)

Query OK, 0 rows affected (0.049 sec)
```